Simultaneous Uplink and Downlink Transmissions for Wireless Powered Communication Networks

Tri-Nhu Do, Van-Dinh Nguyen, Oh-Soon Shin, and Beongku An

***Abstract*—In this letter, we propose a novel simultaneous uplink (UL) and downlink (DL) transmission scheme aiming to achieve a higher spectral efficiency for wireless power communication networks (WPCNs). Specifically, by exploiting full-duplex (FD) radio and wireless power transfer techniques, an FD hybrid access point is able to communicate with a wireless powered UL user and a DL user over the same time-frequency resource. The optimization problem of interest is to maximize the sum-rate of the scheme under a given power budget, which is formulated as a nonconvex problem. To this end, we propose a low-complexity path-following algorithm for its solution. Numerical results reveal that the proposed scheme indeed achieves substantial sum-rate improvement for the WPCNs.**

***Index Terms*—Wireless powered communication networks, full-duplex, sum-rate maximization, path-following algorithm.**

I. Introduction

Full-duplex (FD) radio has been considered as a potential technology to improve the spectral efficiency of wireless communications [1], [2]. Besides, in order to realize the future of self-sustainable wireless devices, radio-frequency energy harvesting (RF-EH) has been emerging as an auspicious method [3], [4]. By adopting RF-EH, the authors in [5] proposed a wireless powered communication network (WPCN) where a hybrid access point (H-AP) first performs wireless power transfer (WPT) to power downlink (DL) users and then processes information received from uplink (UL) users. To improve the spectral efficiency of the prior WPCN, the authors in [6] proposed an FD-WPCN, whose operation in a time slot is carried out in two phases. Specifically, in the first phase, an FD multi-antenna base station (BS) forms an information-energy beam to transfer information and energy simultaneously on DL, while in the second phase, the BS keeps transferring only information on DL while processes the information received on UL.

Differently from existing works, including [6] in particular, in this paper, UL users are equipped with FD radio so that the UL and DL communications of FD-WPCNs can be carried out simultaneously without dividing a time slot into multiple sub-phases. Moreover, we take advantage of the self-interference (SI) introduced by the UL user as a recycling energy resource for itself [7]. In summary, the contributions of the paper are two-fold: (*i*) We propose a novel simultaneous DL and UL transmission scheme for WPCNs, which has not been reported in the literature. This allows UL user to harvest more energy for its UL transmission and exploit the temporal degrees of freedom (DoF) more effectively than [6]. (*ii*) We formulate a sum-rate maximization problem of the considered system, which is intractable since it has a nonconcave objective function subject to nonconvex constraints. Nonetheless, a low-complexity path-following algorithm is developed for its solution. Numerical results show that the proposed scheme offers significant sum-rate improvement as compared to a very recent existing transmission scheme for WPCNs, i.e., [6]. A trade-off between UL and DL rates is also given to provide more insights into the proposed scheme.

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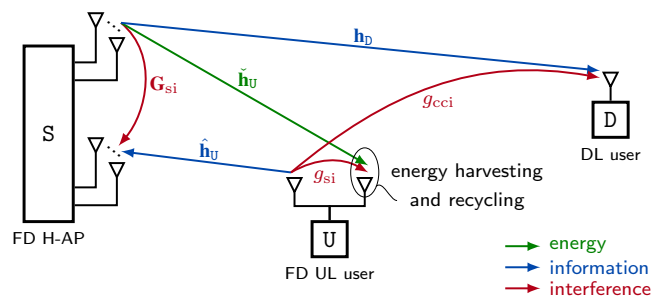

Fig. 1. Illustration of a WPCN with full-duplex uplink transmission.

Notation: Lower-case, bold lower-case, and bold upper-case letters represent scalars, vectors, and matrices, respectively. $(\cdot)^H$, $(\cdot)^T$, and $(\cdot)^*$ denote the Hermitian transpose, normal transpose, and conjugate operations, respectively. $\mathrm{Tr}(\cdot)$ and $\mathbb{E}[\cdot]$ denote the trace and expectation operations, respectively. $|\cdot|$ and $\|\cdot\|_2$ denotes the absolute value and l_2 norm, respectively. $\Re\{\cdot\}$ represents the real part of a complex number.

II. System Model and Problem Formulation

A. System Model

Let us consider a WPCN consisting of a source $\mathtt{S}$, i.e., FD H-AP, one UL user denoted by $\mathtt{U}$ and one DL user denoted by $\mathtt{D}$, as depicted in Fig. 1. The source $\mathtt{S}$ is equipped with N transmit and M receive antennas, and operates in FD mode. In this paper, we consider a worst-case design where the UL user does not have available power to convey its information

symbol to the FD H-AP, and thus needs to harvest energy from the RF signal, as in [6]. To exploit self-energy recycling, we assume that U transmits an information signal to S with one antenna while receiving energy from the other antenna at the same time. DL user is an HD single-antenna user since it intends to receive only the information signal from the FD H-AP. DL and UL transmissions of the proposed WPCN can be described as follows. On the DL transmission, S communicates with D via an information beamforming while concurrently powering U using an energy beamforming. On the UL transmission, U transmits its information signal to S by using the harvested energy. It is noteworthy that DL and UL transmissions are carried out on the same time-frequency resource.

The complex baseband equivalent signal transmitted at the source can be written as $\mathbf{x} = \mathbf{w}x_{\mathtt{D}} + \mathbf{v}x_{\mathtt{e}}$, where $\mathbf{w} \in \mathbb{C}^{N\times 1}$ and $\mathbf{v} \in \mathbb{C}^{N\times 1}$ denote the information and energy beamforming vectors for D and U, respectively, $x_{\mathtt{D}}$ with $\mathbb{E}[|x_{\mathtt{D}}|^2] = 1$ and $x_{\mathtt{e}}$ with $\mathbb{E}[|x_{\mathtt{e}}|^2] = 1$ are the information and energy signals intended for D and U, respectively. Assuming a flat fading channel, the received signals at D and U on the DL channel, respectively, can be expressed as

$$y_{\mathtt{SD}} = \mathbf{h}_{\mathtt{D}}^H \mathbf{w} x_{\mathtt{D}} + \mathbf{h}_{\mathtt{D}}^H \mathbf{v} x_{\mathtt{e}} + p_{\mathtt{U}} g_{\mathrm{cci}} x_{\mathtt{U}} + n_{\mathtt{D}}, \tag{1}$$

$$y_{\mathtt{SU}} = \check{\mathbf{h}}_{\mathtt{U}}^H \mathbf{w} x_{\mathtt{D}} + \check{\mathbf{h}}_{\mathtt{U}}^H \mathbf{v} x_{\mathtt{e}} + p_{\mathtt{U}} \sqrt{\mu_{\mathtt{U}}} g_{\mathrm{si}} x_{\mathtt{U}} + n_{\mathtt{U}}, \tag{2}$$

where $\mathbf{h}_{\mathtt{D}} \in \mathbb{C}^{N\times 1}$ and $\check{\mathbf{h}}_{\mathtt{U}} \in \mathbb{C}^{N\times 1}$ denote the channel vectors from S to D and to U, respectively; $p_{\mathtt{U}}$ and $x_{\mathtt{U}}$ with $\mathbb{E}[|x_{\mathtt{U}}|^2] = 1$ are the transmit power coefficient and message transmitted by U to S; $g_{\mathrm{cci}} \in \mathbb{C}$ denotes a channel coefficient for the co-channel interference (CCI) from U to D; $n_{\mathtt{D}} \sim \mathcal{CN}(0, \sigma_{\mathtt{D}}^2)$ and $n_{\mathtt{U}} \sim \mathcal{CN}(0, \sigma_{\mathtt{U}}^2)$ represent the additive white Gaussian noise (AWGN) at D and U, respectively; g_{si} denotes a channel coefficient for the SI created by the FD transmission at U, while $\mu_{\mathtt{U}} \in [0, 1]$ is used to stand for the degree of SI mitigation after interference isolation and cancellation process at U.

To fully utilize the SI, we further assume that the SI is unsuppressed, i.e., the SI cancellation (SIC) is inactivated, at the UL user. Thus, in the sequel of the paper, we set $\mu_{\mathtt{U}} = 1$, i.e., $\mu_{\mathtt{U}} = 0$ dB, and omit the symbol in equations. The harvested energy at U, i.e., $E_{\mathtt{U}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}) = \eta\mathbb{E}[|y_{\mathtt{SU}}|^2]T$, can be detailed from (2) as $E_{\mathtt{U}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}) = \eta(|\check{\mathbf{h}}_{\mathtt{U}}^H \mathbf{w}|^2 + |\check{\mathbf{h}}_{\mathtt{U}}^H \mathbf{v}|^2 + p_{\mathtt{U}}^2 |g_{\mathrm{si}}|^2)T$, where η and T are the energy conversion efficiency at U and the length of a communication time block, respectively. The term $p_{\mathtt{U}}^2|g_{\mathrm{si}}|^2$ represents the energy recycled from the transmit signal of U. As a result of energy recycling, the UL user is expected to harvest more energy from the RF signal than the counterpart in [6]. Note that the energy recycling model considered in this paper is different from that proposed in [7] since U does not need to use up the harvested energy to transmit its signal. Specifically, the transmit power[1] of U is constrained as described in (4d) in the next subsection, i.e., $p_{\mathtt{U}}^2 \leq P_{\mathtt{U}}^{\mathrm{eh}} \triangleq E_{\mathtt{U}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})/T$.

Meanwhile, UL user U transmits its signal to the source S on the UL channel, which can be expressed as $\mathbf{y}_{\mathtt{US}} = p_{\mathtt{U}}\hat{\mathbf{h}}_{\mathtt{U}} x_{\mathtt{U}} + \sqrt{\mu_{\mathtt{S}}}\mathbf{G}_{\mathrm{si}}^H \mathbf{w} x_{\mathtt{D}} + \sqrt{\mu_{\mathtt{S}}}\mathbf{G}_{\mathrm{si}}^H \mathbf{v} x_{\mathtt{e}} + \mathbf{n}_{\mathtt{S}}$, where $\hat{\mathbf{h}}_{\mathtt{U}} \in \mathbb{C}^{M\times 1}$ represents the UL channel vector from U to S; $\mathbf{G}_{\mathrm{si}} \in \mathbb{C}^{N\times M}$ and $\mu_{\mathtt{S}} \in [0, 1]$ represent the SI channel matrix and the degree of SI mitigation at S, respectively; $\mathbf{n}_{\mathtt{S}} \sim \mathcal{CN}(\mathbf{0}, \sigma_{\mathtt{S}}^2\mathbf{I}_M)$ denotes the AWGN at S.

We assume that the aggregate interferences for both DL and UL are Gaussian distributed. Then, the DL and UL achievable rates in nat/sec/Hz can be written as

$$R_{\mathtt{SD}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}) = \ln\big(1 + \gamma_{\mathtt{SD}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})\big), \tag{3a}$$

$$R_{\mathtt{US}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}) = \ln\big(1 + \gamma_{\mathtt{US}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})\big), \tag{3b}$$

respectively, where $\gamma_{\mathtt{SD}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})$ and $\gamma_{\mathtt{US}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})$ are the signal-to-interference-plus-noise ratio (SINR) at D and S, which are given by $\gamma_{\mathtt{SD}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}) = |\mathbf{h}_{\mathtt{D}}^H \mathbf{w}|^2/(|\mathbf{h}_{\mathtt{D}}^H \mathbf{v}|^2 + p_{\mathtt{U}}^2|g_{\mathrm{cci}}|^2 + \sigma_{\mathtt{D}}^2)$, $\gamma_{\mathtt{US}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}) = p_{\mathtt{U}}^2 \hat{\mathbf{h}}_{\mathtt{U}}^H \mathbf{\Theta}(\mathbf{w}, \mathbf{v})^{-1}\hat{\mathbf{h}}_{\mathtt{U}}$, with $\mathbf{\Theta}(\mathbf{w}, \mathbf{v}) \triangleq \mu_{\mathtt{S}}\mathbf{G}_{\mathrm{si}}^H \mathbf{w}\mathbf{w}^H \mathbf{G}_{\mathrm{si}} + \mu_{\mathtt{S}}\mathbf{G}_{\mathrm{si}}^H \mathbf{v}\mathbf{v}^H \mathbf{G}_{\mathrm{si}} + \sigma_{\mathtt{S}}^2\mathbf{I}_M$. Herein, to maximize the UL rate we adopt the minimum mean square error receiver (MMSE) at the source in decoding the UL signal. Clearly, as seen in (3), both DL and UL transmissions use a full communication time block to transmit their signals, while only a fraction of time block is used for UL transmission in [6]. Therefore, the proposed transmission scheme is expected to achieve better performance.

B. Optimization Problem Formulation

Let $\bar{r}_{\mathtt{D}}$ and $\bar{r}_{\mathtt{U}}$ (nat/sec/Hz) set the rate thresholds for the DL and UL users, respectively; $P_{\mathtt{S}}$ is a given power budget at the H-AP and $P_{\mathtt{U}}^{\mathrm{eh}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}) \triangleq E_{\mathtt{U}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})/T$. The objective of this study is to maximize the sum-rate of the system, which can be mathematically expressed as

$$\underset{\mathbf{w},\mathbf{v},p_{\mathtt{U}}}{\text{maximize}} \quad R_{\mathtt{SD}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}) + R_{\mathtt{US}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}) \tag{4a}$$

$$\text{subject to} \quad R_{\mathtt{SD}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}) \geq \bar{r}_{\mathtt{D}}, \tag{4b}$$

$$R_{\mathtt{US}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}) \geq \bar{r}_{\mathtt{U}}, \tag{4c}$$

$$p_{\mathtt{U}}^2 \leq P_{\mathtt{U}}^{\mathrm{eh}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}), \tag{4d}$$

$$p_{\mathtt{U}} > 0, \tag{4e}$$

$$\|\mathbf{w}\|^2 + \|\mathbf{v}\|^2 \leq P_{\mathtt{S}}. \tag{4f}$$

Since both $R_{\mathtt{SD}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})$ and $R_{\mathtt{US}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})$ are nonconcave functions, problem (4) is nonconvex. Hence, it is challenging to find the optimal solution in general. In the following section, we propose a computationally efficient path-following procedure that leads to a stationary point of (4). Note that the UL and DL users may change their roles in another communication time block, which leads to a different network scenario and the corresponding new optimization problem. Nevertheless, the following mathematical framework of convexification method proposed to solve (4) can still be applied to such a problem.

III. Proposed Path-Following Method

The proposed path-following algorithm is based on the inner approximation framework in [8]. The main idea is to provide a minorant of $R_{\mathtt{SD}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})$ and $R_{\mathtt{US}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})$, which has been widely used in wireless communication designs, e.g., [6], [9].

Let us deal with the nonconcavity of the function $R_{\mathtt{SD}}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})$ first by using the following inequality:

$$f(z) \triangleq \ln(1 + 1/z) \geq \Psi(\bar{z}) - \Xi(\bar{z})z \triangleq \bar{f}(z), \ \forall z, \bar{z}, > 0, \tag{5}$$

[1]We assume that the energy harvested at the UL user can be temporarily stored in the UL user's battery and then is used to transmit data, where the delay is negligible, as in [3], [4].

where $\Psi(\bar{z}) \triangleq \ln(1+1/\bar{z})+1/(\bar{z}+1)$ and $\Xi(\bar{z}) \triangleq 1/\bar{z}(\bar{z}+1)$. Inequality (5) holds true due to the convexity of the function $f(z)$ [9], and the function $\bar{f}(z)$ is a concave minorant of $f(z)$ at $\bar{z}$. Let $1/z := \gamma_{\mathtt{SD}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}})$ and $1/\bar{z} = 1/z^{(k)} := \gamma_{\mathtt{SD}}(\mathbf{w}^{(k)},\mathbf{v}^{(k)},p_{\mathtt{U}}^{(k)})$, where $(\mathbf{w}^{(k)},\mathbf{v}^{(k)},p_{\mathtt{U}}^{(k)})$ is a feasible point at the $(k-1)$-th iteration. Then, it follows that

$$R_{\mathtt{SD}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) \geq \Psi(z^{(k)}) - \Xi(z^{(k)})\frac{|\mathbf{h}_{\mathtt{D}}^H\mathbf{v}|^2+p_{\mathtt{U}}^2|g_{\mathrm{cci}}|^2+\sigma_{\mathtt{D}}^2}{|\mathbf{h}_{\mathtt{D}}^H\mathbf{w}|^2}. \quad (6)$$

In addition, we can equivalently replace $|\mathbf{h}_{\mathtt{D}}^H\mathbf{w}|^2$ by $(\Re\{\mathbf{h}_{\mathtt{D}}^H\bar{\mathbf{w}}\})^2$ with $\Re\{\mathbf{h}_{\mathtt{D}}^H\bar{\mathbf{w}}\} \geq 0$ [9]. Due to the convexity of function $|t|^2$, the following inequality holds:

$$|t|^2 \geq 2(t^{(k)})^* t - |t^{(k)}|^2,\ \forall t,\ t^{(k)} \in \mathbb{C}. \quad (7)$$

From (6) and (7), a concave quadratic minorant of $R_{\mathtt{SD}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}})$ at $(\mathbf{w}^{(k)},\mathbf{v}^{(k)},p_{\mathtt{U}}^{(k)})$ reads as

$$\mathcal{R}_{\mathtt{SD}}^{(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) := \Psi(z^{(k)}) - \Xi(z^{(k)})\frac{|\mathbf{h}_{\mathtt{D}}^H\mathbf{v}|^2+p_{\mathtt{U}}^2|g_{\mathrm{cci}}|^2+\sigma_{\mathtt{D}}^2}{\Phi^{(k)}(\mathbf{w})} \quad (8)$$

over the trust region

$$\Phi^{(k)}(\mathbf{w}) \triangleq \Re\{\mathbf{h}_{\mathtt{D}}^H\mathbf{w}^{(k)}\}(2\Re\{\mathbf{h}_{\mathtt{D}}^H\mathbf{w}\} - \Re\{\mathbf{h}_{\mathtt{D}}^H\mathbf{w}^{(k)}\}) > 0. \quad (9)$$

Next, we turn our attention to the nonconcave function $R_{\mathtt{US}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}})$. It is clear that the steps used for tackling $R_{\mathtt{SD}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}})$ cannot be extended to $R_{\mathtt{US}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}})$ since the approximate function resulting from (5) is still highly nonconcave. At this point, we resort on the following inequality [6, Eq. (20)]:

$$\begin{aligned}\ln\left(1+|x|^2/y\right) \geq \ln\left(1+|\bar{x}|^2/\bar{y}\right) - |\bar{x}|^2/\bar{y} + 2\Re\{\bar{x}^* x\}/\bar{y} \\ - \left[\bar{y}^{-1} - (|\bar{x}|^2+\bar{y})^{-1}\right](|x|^2+y), \\ \forall x \in \mathbb{C}, \bar{x} \in \mathbb{C}, y > 0, \bar{y} > 0. \quad (10)\end{aligned}$$

Applying (10) to $R_{\mathtt{US}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}})$, the concave quadratic minorant of $R_{\mathtt{US}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}})$ at $(\mathbf{w}^{(k)},\mathbf{v}^{(k)},p_{\mathtt{U}}^{(k)})$ is given by

$$\begin{aligned}R_{\mathtt{US}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) &\geq \alpha^{(k)} + \beta^{(k)}p_{\mathtt{U}} - \varphi(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) \quad (11)\\ &:= \mathcal{R}_{\mathtt{US}}^{(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}),\end{aligned}$$

where
$\alpha^{(k)} \triangleq \ln\big(1+\gamma_{\mathtt{US}}(\mathbf{w}^{(k)},\mathbf{v}^{(k)},p_{\mathtt{U}}^{(k)})\big) - \gamma_{\mathtt{US}}(\mathbf{w}^{(k)},\mathbf{v}^{(k)},p_{\mathtt{U}}^{(k)})$,
$\beta^{(k)} \triangleq 2p_{\mathtt{U}}^{(k)}\hat{\mathbf{h}}_{\mathtt{U}}^H\mathbf{\Theta}(\mathbf{w}^{(k)},\mathbf{v}^{(k)})^{-1}\hat{\mathbf{h}}_{\mathtt{U}}$,
$\varphi(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) \triangleq \mathrm{Tr}\big([\mathbf{\Theta}(\mathbf{w}^{(k)},\mathbf{v}^{(k)})^{-1} - (\mathbf{\Theta}(\mathbf{w}^{(k)},\mathbf{v}^{(k)}) + (p_{\mathtt{U}}^{(k)})^2\hat{\mathbf{h}}_{\mathtt{U}}\hat{\mathbf{h}}_{\mathtt{U}}^H)^{-1}](\mathbf{\Theta}(\mathbf{w},\mathbf{v}) + p_{\mathtt{U}}^2\hat{\mathbf{h}}_{\mathtt{U}}\hat{\mathbf{h}}_{\mathtt{U}}^H)\big)$.
As a result, constraints (4b) and (4c) are innerly approximated by the following convex quadratic constraints

$$\mathcal{R}_{\mathtt{SD}}^{(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) \geq \bar{r}_{\mathtt{D}}, \quad (12)$$

$$\mathcal{R}_{\mathtt{US}}^{(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) \geq \bar{r}_{\mathtt{U}}. \quad (13)$$

For constraint (4d), it can be observed that both sides are quadratic functions, which are also convex functions. Naturally, we can apply the inner approximation method to expose the hidden convexity of (4d). Specifically, we iteratively replace (4d) by

$$p_{\mathtt{U}}^2 \leq \mathcal{P}_{\mathtt{U}}^{\mathrm{eh},(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}), \quad (14)$$

where $\mathcal{P}_{\mathtt{U}}^{\mathrm{eh},(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}})$ is the global lower bound of $P_{\mathtt{U}}^{\mathrm{eh}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}})$ at the feasible point $(\mathbf{w}^{(k)},\mathbf{v}^{(k)},p_{\mathtt{U}}^{(k)})$ and it is given as

$$\begin{aligned}\mathcal{P}_{\mathtt{U}}^{\mathrm{eh},(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) := \eta\big[&2\Re\{(\mathbf{w}^{(k)})^H\check{\mathbf{h}}_{\mathtt{U}}\check{\mathbf{h}}_{\mathtt{U}}^H\mathbf{w}\} - |\check{\mathbf{h}}_{\mathtt{U}}^H\mathbf{w}^{(k)}|^2 \\ &+ 2\Re\{(\mathbf{v}^{(k)})^H\check{\mathbf{h}}_{\mathtt{U}}\check{\mathbf{h}}_{\mathtt{U}}^H\mathbf{v}\} - |\check{\mathbf{h}}_{\mathtt{U}}^H\mathbf{v}^{(k)}|^2 \\ &+ |g_{\mathrm{si}}|^2(2p_{\mathtt{U}}^{(k)}p_{\mathtt{U}} - (p_{\mathtt{U}}^{(k)})^2)\big],\end{aligned}$$

by using (7). In summary, the following convex quadratic program, which provides a minorant maximization for (15), is solved at the k-th iteration:

$$\underset{\mathbf{w},\mathbf{v},p_{\mathtt{U}}}{\text{maximize}} \quad \psi^{(k)} \triangleq \mathcal{R}_{\mathtt{SD}}^{(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) + \mathcal{R}_{\mathtt{US}}^{(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) \quad (15a)$$

$$\text{subject to} \quad (4e), (4f), (9), (12), (13), (14). \quad (15b)$$

The proposed path-following procedure used to solve the sum-rate optimization problem (4) is summarized in Algorithm 1.

Algorithm 1 Proposed Path-following Algorithm to Solve (4)

Initialization:
1: Solve (16) to generate an initial feasible point.
2: Set $k := 0$.

Path-following procedure:
3: **repeat**
4: Solve (15) with $(\mathbf{w}^{(k)},\mathbf{v}^{(k)},p_{\mathtt{U}}^{(k)})$ to obtain the optimal solution $(\mathbf{w}^\star,\mathbf{v}^\star,p_{\mathtt{U}}^\star)$.
5: Update $\mathbf{w}^{(k+1)} := \mathbf{w}^\star, \mathbf{v}^{(k+1)} := \mathbf{v}^\star, p_{\mathtt{U}}^{(k+1)} := p_{\mathtt{U}}^\star$.
6: Set $k := k+1$.
7: **until** Convergence

Generation of an Initial Point: To successfully initialize the computational procedure, we now present a variant of Algorithm 1 as follows. We start with modifying (4) into the following problem of Quality-of-Service (QoS) feasibility: $\underset{\mathbf{w},\mathbf{v},p_{\mathtt{U}}}{\text{maximize}} \min\ \{R_{\mathtt{SD}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) - \bar{r}_{\mathtt{D}}, R_{\mathtt{US}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) - \bar{r}_{\mathtt{U}}, P_{\mathtt{U}}^{\mathrm{eh}}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) - p_{\mathtt{U}}^2\}$, subject to (4e), (4f), (9).

Next, we randomly generate $p_{\mathtt{U}} > 0$ and $(\mathbf{w},\mathbf{v})$ over $\mathbb{C}^{N\times 1}$, and then scale $(\mathbf{w},\mathbf{v})$ properly to satisfy (4f) and (9). Then, the following convex program is invoked:

$$\underset{\mathbf{w},\mathbf{v},p_{\mathtt{U}}}{\text{maximize}}\ \phi, \quad \text{subject to} \quad (4e), (4f), (9), \quad (16)$$

where $\phi \triangleq \min\{\mathcal{R}_{\mathtt{SD}}^{(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) - \bar{r}_{\mathtt{D}},\ \mathcal{R}_{\mathtt{US}}^{(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) - \bar{r}_{\mathtt{U}},\ \mathcal{P}_{\mathtt{U}}^{\mathrm{eh},(k)}(\mathbf{w},\mathbf{v},p_{\mathtt{U}}) - p_{\mathtt{U}}^2\}$. The solution of (16) can provide an initial feasible point to execute Algorithm 1 whenever $\phi \geq 0$.

Convergence Analysis: The transformation of nonconcave objective function (4a) and nonconvex constraints (4b), (4c), (4d) satisfy the properties listed in [8], i.e., the feasible point of the transformed function is also guaranteed to be feasible for the original function. Therefore, the proposed path-following procedure to solve (4) generates a non-decreasing sequence of objective values, i.e. $\psi^{(k+1)} \geq \psi^{(k)}$ that provably converges to a limited point due to (4f). One can prove that, as in [8]–[10], the sequence of the obtained solutions for (4) converges at least to a locally optimal solution satisfying the Karush-Kuhn-Tucker (KKT) optimality conditions.

Complexity Comparison: As problem (15) has $2N+1$ real-valued scalar decision variables and 6 quadratic and linear constraints, its computational complexity per iteration is $\mathcal{O}\big(6^{2.5}(4N^2+4N+7)\big)$. On the other hand, the per-iteration complexity of solving (48) in [6] for two-user WPCN system is $\mathcal{O}\big(7^{2.5}(9N^2+4M^2+12MN+24N+16M+23)\big)$, which is indeed higher than that of the proposed convex quadratic

problem (15).

Application of a Conic Programming: As an effort to reduce the computation time, we further formulate (15) as a second-order cone (SOC) programming which can be solved more efficiently by modern convex solvers. To arrive at the SOC representation, we use the fact that a hyperbolic constraint $z^2 \leq xy$ for $x, y \geq 0$ is equivalent to $\|[z \quad 0.5(x-y)]^T\|_2 \leq 0.5(x+y)$. In doing so, we introduce slack variables $\vartheta \geq 0$ and $\omega \geq 0$ to equivalently rewrite $\mathcal{R}_{\mathtt{SD}}^{(k)}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})$ and $\mathcal{R}_{\mathtt{US}}^{(k)}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})$ by $\mathcal{R}_{\mathtt{SD}}^{(k)}(\vartheta) := \Psi(z^{(k)}) - \Xi(z^{(k)})\vartheta$ and $\mathcal{R}_{\mathtt{US}}^{(k)}(p_{\mathtt{U}}, \omega) := \alpha^{(k)} + \beta^{(k)} p_{\mathtt{U}} - \omega$, with imposing the following SOC constraints:

$$\left\| \left[\mathbf{h}_{\mathtt{D}}^H \mathbf{v}, p_{\mathtt{U}} g_{\mathrm{cci}}, \sigma_{\mathtt{D}}, 0.5(\Phi^{(k)}(\mathbf{w}) - \vartheta)\right]^T \right\|_2 \leq 0.5(\Phi^{(k)}(\mathbf{w}) + \vartheta), \tag{17}$$

$$\left\| \left[\sqrt{\mu_{\mathtt{S}}} \mathbf{w}^H \mathbf{G}_{\mathrm{si}} \bar{\mathbf{\Theta}}^{0.5}, \sqrt{\mu_{\mathtt{S}}} \mathbf{v}^H \mathbf{G}_{\mathrm{si}} \bar{\mathbf{\Theta}}^{0.5}, \sigma_{\mathtt{S}} \mathrm{Tr}(\bar{\mathbf{\Theta}})^{0.5}, \right.\right.$$
$$\left.\left. p_{\mathtt{U}} \hat{\mathbf{h}}_{\mathtt{U}}^H \bar{\mathbf{\Theta}}^{0.5}, 0.5(\omega - 1)\right]^T \right\|_2 \leq 0.5(\omega + 1), \tag{18}$$

where

$$\bar{\mathbf{\Theta}} \triangleq \mathbf{\Theta}(\mathbf{w}^{(k)}, \mathbf{v}^{(k)})^{-1} - \left(\mathbf{\Theta}(\mathbf{w}^{(k)}, \mathbf{v}^{(k)}) + (p_{\mathtt{U}}^{(k)})^2 \hat{\mathbf{h}}_{\mathtt{U}} \hat{\mathbf{h}}_{\mathtt{U}}^H\right)^{-1}.$$

Since constraints (4f) and (14) admit SOC representation, we reformulate problem (15) as the following SOC programming:

$$\underset{\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}, \vartheta, \omega}{\text{maximize}} \quad \psi^{(k)} \triangleq \mathcal{R}_{\mathtt{SD}}^{(k)}(\vartheta) + \mathcal{R}_{\mathtt{US}}^{(k)}(p_{\mathtt{U}}, \omega) \tag{19a}$$

$$\text{subject to} \quad \mathcal{R}_{\mathtt{SD}}^{(k)}(\vartheta) \geq \bar{r}_{\mathtt{D}}, \mathcal{R}_{\mathtt{US}}^{(k)}(p_{\mathtt{U}}, \omega) \geq \bar{r}_{\mathtt{U}},$$
$$(4e), (4f), (9), (14), (17), (18). \tag{19b}$$

The equivalence between (15) and (19) is guaranteed by noting the fact that constraints (17) and (18) must hold with equality at optimum. Note that we consider a simplified system to facilitate the performance evaluation of the proposed scheme. Nonetheless, for a large-scale network, the proposed mathematical framework of convexification method can still be applied to the corresponding optimization problem.

IV. Numerical Results

For the purpose of exposition, the simulation conditions are set up as follows. The channels $(\mathbf{h}_{\mathtt{D}}, \check{\mathbf{h}}_{\mathtt{U}}, \hat{\mathbf{h}}_{\mathtt{U}}, g_{\mathrm{cci}})$ are assumed to follow complex Gaussian distribution with zero mean, as in [6]. The variance of $\mathbf{h}_{\mathtt{D}}$ is set to -30 dB while that of $(\check{\mathbf{h}}_{\mathtt{U}}, \hat{\mathbf{h}}_{\mathtt{U}})$ to -20 dB, which implies that the UL user should be deployed closer to the H-AP than the DL user due to the practical constraint of the RF-EH. The variance of the CCI channel g_{cci} is set to -20 dB. The SI channels $(\mathbf{G}_{\mathrm{si}}, g_{\mathrm{si}})$ are generated as independent and identically distributed Rician random variables with unit non-centrality parameter and the scale parameter. Other parameters are set as $M = N = 3$, $\sigma_{\mathtt{S}}^2 = \sigma_{\mathtt{D}}^2 = \sigma_{\mathtt{U}}^2 = -80$ dBm, the signal bandwidth is 1 MHz.

In Fig. 2, we provide the performance comparison of the proposed scheme with and without energy beamforming (EB), and the scheme proposed in [6], where the sum-rate is plotted as a function of the transmit power budget under different values of η. As can be observed, the proposed scheme indeed provides better performance than the prior work in [6]. Specifically, when $\eta = 0.5$, the use of the EB shows its effectiveness since the proposed scheme can achieve much higher sum-rate than that in [6]. The sum-rates of all considered schemes significantly decrease as the energy conversion efficiency drops from 0.5 to 0.1. Also, when $\eta = 0.1$, the performance gaps between the schemes are not significant. Additionally, according to our study, the power allocated for the EB is around 0.7% to 1% of the total effective transmit power budget.

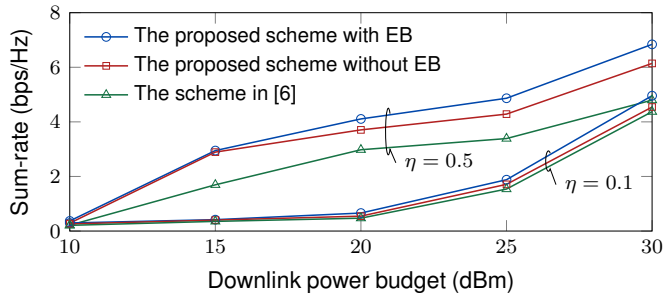

Fig. 2. Sum-rate comparison between the proposed scheme with and without EB, and the scheme in [6] with $\mu_{\mathtt{S}} = -110$ dB and $\bar{r}_{\mathtt{U}} = 1$ (bps/Hz).

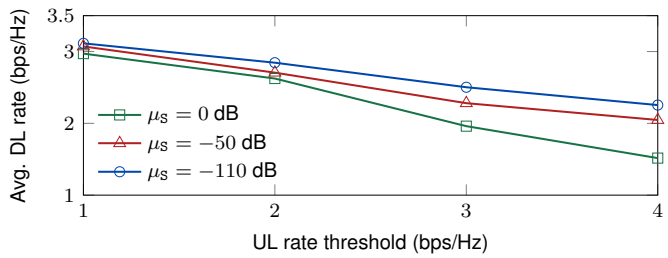

Fig. 3. Trade-off between UL and DL rates under different values of the degree of SI mitigation at the H-AP with $\eta = 0.5$ and $P_{\mathtt{S}} = 25$ dBm.

In Fig. 3, we study the impact of the UL data rate threshold and the degree of SI mitigation on the achievable average DL data rate, which is obtained by reformulating problem (15) as $\underset{\mathbf{w}, \mathbf{v}, p_{\mathtt{U}}}{\text{maximize}}\ \mathcal{R}_{\mathtt{SD}}^{(k)}(\mathbf{w}, \mathbf{v}, p_{\mathtt{U}})$, subject to (4e), (4f), (9), (12), (14). As can be seen, for a given power budget and a given UL data rate threshold, the average DL rate decreases as $\mu_{\mathtt{S}}$ increases since there is a stronger SI at the H-AP. In addition, the average DL data rate decreases as the QoS of the UL transmission increases. Moreover, we numerically observe that when $P_{\mathtt{S}}$ is low, the recycled energy highly contributes to the transmit power of the UL user; in contrast, when $P_{\mathtt{S}}$ is high, the role of the energy recycling at the UL user becomes less significant.

V. Conclusion

In this paper, we have proposed a novel simultaneous UL and DL transmission scheme for the WPCNs. Specifically, by adopting WPT and FD techniques, the UL user can harvest energy from the H-AP and recycle energy from the SI. We have formulated the nonconvex optimization problem of the sum-rate maximization and proposed the path-following algorithm to attain at least a locally optimal solution. Numerical results have shown that the proposed scheme provides essential performance improvements for the WPCNs.

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